

Drought tolerance as an evolutionary precursor to frost and winter tolerance in grasses

Laura Schat^{1,2}, Marian Schubert^{1,2,§}, Siri Fjellheim³, Aelys M. Humphreys^{1,2}

¹Department of Ecology, Environment and Plant Sciences, Stockholm University, Stockholm, Sweden

²Bolin Centre for Climate Research, Stockholm University, Stockholm, Sweden

³Department of Plant Sciences, Norwegian University of Life Sciences, Ås, Norway

[§]Present address: Department of Plant Sciences, Norwegian University of Life Sciences, Ås, Norway

Corresponding authors: Department of Ecology, Environment and Plant Sciences, Stockholm University, Stockholm, Sweden. Email: aelys.humphreys@su.se; Department of Ecology, Environment and Plant Sciences, Stockholm University, Stockholm, Sweden. Email: laura.schat@su.se

Abstract

Accumulating evidence is suggesting more frequent tropical-to-temperate transitions than previously thought. This raises the possibility that biome transitions could be facilitated by precursor traits. A wealth of ecological, genetic, and physiological evidence suggests overlap between drought and frost stress responses, but the origin of this overlap, i.e., the evolution of these responses relative to each other, is poorly known. Here, we test whether adaptation to frost and/or severe winters in grasses (Poaceae) was facilitated by ancestral adaptation to drought. We used occurrence patterns across Köppen–Geiger climate zones to classify species as drought, frost, and/or winter tolerant, followed by comparative analyses. Ancestral state reconstructions revealed different evolutionary trajectories in different clades, suggesting both drought-first and frost-first scenarios. Explicit simultaneous modelling of drought and frost/winter tolerance provided some support for correlated evolution, but suggested higher rates of gain of frost/winter tolerance in drought-sensitive rather than drought-tolerant lineages. Overall, there is limited support across grasses as a whole that drought tolerance acted as an evolutionary precursor to frost or severe winter tolerance. Different scenarios in different clades is consistent with present-day grasses being either cold or drought specialists, possibly as a consequence of trade-offs between different stress tolerance responses.

Keywords: biome transition, correlated evolution, GBIF data, Köppen–Geiger, rate heterogeneity, simulation

Introduction

A general pattern of plant diversity is that it is highest in the warm, wet tropics and declines toward the poles, where conditions become colder and more seasonally variable. Most major angiosperm clades originated in the tropics, as tropical climates predate temperate ones (e.g., Judd et al., 1994; Ricklefs & Renner, 1994; Zanne et al., 2014). When temperate biomes expanded after the Eocene, tropical lineages in emerging temperate biomes had to adapt to drier, colder, and more seasonal climates to persist (Eldrett et al., 2009; Kerkhoff et al., 2014; Zachos et al., 2001). Yet, many clades do not occur at higher latitudes, despite plenty of evolutionary opportunity and time to become established there (Donoghue, 2008). Causes of this latitudinal diversity gradient have been the subject of a debate spanning decades (Fine & Ree, 2006; Hillebrand, 2004; Jansson et al., 2013; Kerkhoff et al., 2014; Mittelbach et al., 2007; Rohde, 1992; Wiens & Donoghue, 2004; Wiens & Graham, 2005). One prevalent explanation assumes that tropical-to-temperate transitions have been infrequent because adapting to novel, and especially freezing, climates is difficult (e.g., Wiens & Donoghue, 2004; Wiens & Graham, 2005; Donoghue, 2008; Wiens et al., 2010; Körner, 2016). As a result, temperate clades are expected to be clustered phylogenetically and nested within tropical clades (Kerkhoff et al., 2014).

Previous reviews have suggested that approximately 40%–50% of all angiosperm families have temperate lineages (Preston & Sandve, 2013; Stevens, 2001), and indeed, on a global scale, biome shifts are thought to have been rare (Crisp et al., 2009). However, recent research is indicating that tropical-to-temperate transitions may have been more frequent than currently thought (Donoghue & Edwards, 2014; Jansson et al., 2013; Nürk et al., 2018; Zizka et al., 2020), apparently having occurred multiple times independently, even within single families (e.g., Poaceae; Edwards & Smith, 2010; Schubert et al., 2020; Amaranthaceae; Cousins-Westerberg et al., 2023) and genera (e.g., *Viburnum*; Schmerler et al., 2012; Spriggs et al., 2015). This raises the possibility that such transitions are not necessarily as difficult as widely assumed, which in turn could imply that biome transitions are facilitated by evolutionary precursor trait(s) (also referred to as “pre-adaptations” or “enabler traits”; Donoghue & Edwards, 2014; Edwards & Donoghue, 2013). For tropical-to-temperate transitions, such precursors could be related to ancient stress tolerance mechanisms that have been repurposed for the specific challenges of a cold, temperate climate (Preston & Sandve, 2013). If so, we would expect adaptations to frost and long winters to have arisen more frequently in clades with such evolutionary precursor(s) than in clades without them, leading to a pattern of multiple apparent “independent” transitions to a temperate climate in those clades.

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Temperate plants are confronted by a range of abiotic stresses. For example, frost impacts plants on many levels, ranging from dehydration stress caused by the lack of fluid water to damage to the structural integrity of cells and biomembranes caused by ice crystal formation (Preston & Sandve, 2013; Sakai & Larcher, 1987). In addition, plant growth slows down at temperatures below 5 °C and comes to a near standstill at freezing (Körner, 2016; Nievola et al., 2017). Overcoming these abiotic stresses and evolving frost tolerance requires major physiological innovations (Donoghue, 2008; Lancaster & Humphreys, 2020; Sakai & Larcher, 1987; Wiens & Donoghue, 2004).

Plants in freezing areas can be exposed to frost episodically (e.g., diurnally), or periodically (i.e., during prolonged periods of seasonal or even permanent frost). Temperatures can drop below freezing occasionally in the tropics (e.g., on tropical mountains) and across the temperate zone, but plants in areas with long, cold winters are in addition confronted by seasonal variation in precipitation, temperature and day length and must endure long periods of low (but not necessarily freezing) temperatures. Frost tolerance alone is thus insufficient to survive cold, temperate winters, which requires a suite of physiological and phenological adjustments in response to seasonal change (cold acclimation and vernalization) and a short (and possibly cold) growing season. To capture some of the stress tolerance mechanisms required by temperate plants, we separately consider exposure to both frost and severe winters in this study.

There is strong ecological, genetic, and physiological evidence that adaptations to low temperature and drought stress are not independent. On a cellular level, the dehydration caused by extracellular ice is virtually identical to dehydration stress caused by drought (Körner, 2016; Sakai & Larcher, 1987), and several genes involved in responses to cold and frost stress are also expressed during drought stress (Zhong et al., 2018; Schubert et al., 2019; Birkeland et al., 2022; Das et al., 2023). Both periodic drought and winter present long periods without possibilities for photosynthesis, and plants need storage carbohydrates that can be hydrolyzed to provide fuel to survive both of these conditions. Furthermore, plants from dry mountains are more frost tolerant than those from wet ones (Sierra-Almeida et al., 2016), and experimental water-deficiency pre-treatment can increase frost tolerance, whereas pre-treatment with heat does not affect subsequent frost tolerance (Sumner et al., 2022).

However, while there is plenty of evidence for a correlation between responses to frost and drought stress, much less is known about how the overlap among different abiotic stress responses evolved. One hypothesis is that ancient stress tolerance mechanisms might have been beneficial to surviving in both cold and dry areas (Folk et al., 2020; Preston & Sandve, 2013). Another hypothesis is that lineages that transitioned to freezing areas might have been predisposed to do so thanks to precursor traits that are beneficial in a freezing environment but evolved in response to aridity in the tropics (e.g., small conduits and a herbaceous habit; Zanne et al., 2014).

In general, some strategy for avoiding dehydration is thought to be more ancient than any adaptation to low temperature stress. Early land plants (Embryophyta) transitioning to terrestrial environments ca. 500 million years (Ma) ago were most likely desiccation tolerant; thus desiccation tolerance is thought to be an ancestral trait in land plants and a key component of adaptation to life on land (Bowles

et al., 2021; Oliver, 1917; Preston & Sandve, 2013; Sakai & Larcher, 1987). However, desiccation tolerance is less common in vascular plants (Tracheophyta); their origin saw the evolution of vascular tissue and other regulatory and morphological complexities that enabled them to tolerate dry conditions (Bowles et al., 2021; Harrison, 2017). Thus, desiccation tolerance was probably lost in the ancestor of vascular plants and replaced by early forms of drought tolerance (Bowles et al., 2021). In contrast, while cool climate pockets may have been present in mid-latitude mountain areas in the Northern Hemisphere in the Eocene (56–34 Ma; Hagen et al., 2019), the emergence of the cold and freezing environments of today is not thought to have begun until the late Eocene or early Oligocene (i.e., mainly in the past ca. 34 Ma; Eldrett et al., 2009; Liu et al., 2009; Pound & Salzmann, 2017; Zachos et al., 2001). Thus, strategies for dealing with low temperature stress that are shared across living land plants are thought to have evolved by independent repurposing of ancestral stress pathways (e.g., related to dehydration; Preston & Sandve, 2013; Schubert et al., 2020; Schubert, Grønvold & et al., 2019), rather than representing ancestral traits.

Here we explore the idea of existing stress responses facilitating adaptation to freezing environments in grasses (Poaceae), by testing whether there is evidence that drought tolerance could have acted as the evolutionary precursor that facilitated persistence in the emerging cold environments of the Oligocene. Poaceae is one of the five largest angiosperm families, with over 11,000 species and 800 genera distributed largely in two main clades: the BOP clade (Bambusoideae, Oryzoideae, and Pooideae) and PACMAD clade (Panicoideae, Aristidoideae, Chloridoideae, Micrairoideae, Arundinoideae, and Danthonioideae; Kellogg, 2001; Clayton et al., 2006; Hodkinson, 2018; Figure 1 and 2). They are well suited for testing our hypothesis of precursor traits due to their cosmopolitan distribution, occurring from moist tropical forests and tropical savannahs to arctic tundras and alpine meadows, including some of the harshest environments on Earth (e.g., hot and cold deserts, saline environments, and geothermal soils; Bennett et al., 2013; Kellogg, 2015; Linder et al., 2018; Nuppenau et al., 2023; Strömberg, 2011; Visser et al., 2013).

Grasses are thought to have originated in the Cretaceous (ca. 93–125 Ma ago; Gallaher et al., 2019; Huang et al., 2016; Schubert et al., 2020), most likely in deep shade, as understory plants of moist, warm forests (Bouchenak-Khelladi et al., 2010; Gallaher et al., 2019; Kellogg, 2001; Strömberg, 2011). Such habitats are today characteristic of a few small lineages that are successively sister to the rest of the family (Anomochloideae, Pharoideae, and Puellioideae), the bamboos (Bambusoideae) and *Brachyelytrum*, which is sister to the rest of the Pooideae. However, the vast majority of grasses today occupy open environments (Elliott et al., 2023).

Transitions from closed to open environments are thought to have happened multiple times, certainly independently in the BOP and PACMAD clades, but exactly how many times and in which lineages is still unclear. Estimates range from one to as many as four independent transitions in the BOP clade, mainly in Pooideae, and one or possibly two in the PACMAD clade (Bouchenak-Khelladi et al., 2010; Elliott et al., 2023; Kellogg, 2001; Zhang et al., 2022a). Both fossil and phylogenetic evidence suggests grasses began transitioning to open habitats around 67–58 Ma (Gallaher et al., 2019; Schubert, Marcussen & et al., 2019; Strömberg, 2011), at different times on different continents (Edwards et al., 2010;

Strömberg, 2011). These transitions most likely occurred in response to increased aridity combined with altered disturbance regimes (seasonality, fire, and herbivory), with less support for a role for global cooling (Edwards et al., 2010; Kellogg, 2015; Strömberg, 2011).

Similarly, grasses have transitioned from tropical to temperate climates multiple times independently, most notably leading to the radiation of two distantly related lineages into cool temperate zones, the Pooideae and Danthonioideae. However, frost tolerance has never been reconstructed as deeply as the nodes at which transitions to open habitats are implied (Edwards & Smith, 2010; Sandve & Fjellheim, 2010; Humphreys & Linder, 2013; Linder et al., 2018; Schubert et al., 2019; Elliott et al., 2023). This suggests that grasses adapted to aridity before cold and frost and sets up predictions for which clades may have been predisposed to adapting to low temperatures under the hypothesis tested here. The level of cold and drought experienced by present-day grasses varies tremendously, e.g., with certain bamboos and panicoid grasses also occurring in cold climates and cold temperate grasses occurring in both mesic (e.g., *Glyceria*) and dry (e.g., *Triticum*) habitats (Edwards & Smith, 2010; Humphreys & Linder, 2013; Schubert, Marcussen & et al., 2019; Schubert et al., 2020; Visser et al., 2013; Watcharamongkol et al., 2018; Zhang et al., 2022a). Thus, the hypothesis that drought tolerance was a precursor trait to frost tolerance requires detailed analysis.

We use occurrence in areas that experience drought, frost, and severe winter, according to the Köppen–Geiger climate classification (Beck et al., 2018; Köppen, 2011), as a proxy for tolerance of each of these stresses. We refer to these stress tolerances in their broadest sense, e.g., “frost tolerance” includes frost escape (e.g., spring annuals), frost avoidance or resistance (e.g., supercooling), and true frost tolerance (e.g., control of ice crystal formation). We use phylogenetic comparative methods to test for evidence consistent with drought tolerance as a precursor for adapting to a temperate climate, specifically, to frost and severe winters. We first reconstruct ancestral states for drought, frost, and winter tolerance separately to visually assess whether (1) frost- and/or winter-tolerant clades are nested within drought-tolerant clades, or appear to arise more frequently in clades that were ancestrally drought tolerant. Next, because evolutionary transitions happen along branches and not at nodes, we model evolutionary transitions between sensitive and tolerant states simultaneously for two traits to test whether (2) drought tolerance evolves first and transitions to frost and winter tolerance are more frequent in drought-tolerant than drought-sensitive lineages. Finally, we (3) verify the models of correlated evolution with alternative null models and simulated data, and (4) test the effect of different species distribution thresholds for scoring species as drought, frost, or winter tolerant.

Methods

Geographical and phylogenetic data

Geographical data were obtained from the Global Biodiversity Information Facility (GBIF, www.gbif.org). All GBIF entries with the family name “Poaceae” were downloaded on 25-02-2020 (<https://doi.org/10.15468/dl.26b2o8>), resulting in over 28 million observations. To remove erroneous and inaccurate observations, the following filtering steps were then followed. Observations that had coordinates with fewer than two

decimals in either the latitude or longitude were removed. The CoordinateCleaner package (v. 2.0-15; Zizka et al., 2019) in R (v 3.6.3; R Core Team, 2020) was used to filter out observations with invalid coordinates, with equal or zero latitude and longitude coordinates, observations located in oceans, biodiversity institutions, GBIF headquarters, capital cities, country or province centroids, and observations where the coordinates did not match the country code. Several major crop and horticultural species were removed manually, as these are likely to occur far outside their native ranges (and potentially climate niches). These were: wheat (*Triticum aestivum* L.), barley (*Hordeum vulgare* L.), sugarcane (*Saccharum officinarum* L.), corn (*Zea mays* L.), oats (*Avena sativa* L.), rice (*Oryza sativa* L.), sorghum (*Sorghum bicolor* (L.) Moench), rye (*Secale cereal* L.), ryegrasses (*Lolium multiflorum* Lam. and *Lolium perenne* L.), silvergrasses (*Miscanthus sinensis* Andersson and *Miscanthus giganteus* J.M.Greef & Deuter ex Hodk. & Renvoize), pampas grass (*Cortaderia selloana* (Schult. & Schult.f.) Asch. & Graebn.), switchgrass (*Panicum virgatum* L.), and several millets (*Setaria italica* (L.) P. Beauv., *Eleusine coracana* (L.) Gaertn., *Cenchrus americanus* (L.) Morrone, and *Panicum dichotomiflorum* Michx.).

Next, the geographical observations were matched to a phylogenetic tree comprising of 3,595 species (Spriggs et al., 2014), using both the currently accepted species name (“species”) and the name used in the original GBIF entry (“verbatimScientificName”). This was done to minimize the chance of omitting species due to synonymy differences (e.g., if two species were previously considered separate but have now been synonymized, and are represented by two tips in the tree). We chose this method as currently synonymous tips were not always each other’s closest relatives, and we did not want to make (arbitrary) decisions on pruning one of these tips. This approach also opens the possibility of a single GBIF observation being matched to two different tips in the tree; however, this was true for only a tiny fraction of all data: in the final dataset 19,924 observations (0.33%) were matched to two species (tips). Overall, 2,531 species with GBIF data could be matched to the tree using currently accepted names and 2,546 using the name in the original GBIF entry. In the end, the filtering and species matching steps resulted in a dataset comprising of phylogenetic information and at least ten GBIF observations for 2,731 species.

As part of our filtering process, we deemed geographical information based on fewer than ten GBIF observations per species unreliable and those species were therefore removed from the dataset (428 species). However, low taxon sampling may be at least as big of a limitation as few geographical occurrence records, and recent studies have included fewer or less precise geographical occurrence data (Pironon et al., 2024; Smith et al., 2023). To include as many species as possible, we therefore retained species with 5–9 GBIF observations if they are also known only from a single botanical country or first-level province, as defined under the World Geographical Scheme for Recording Plant Distributions by the Taxonomic Databases Working Group (TDWG; Brummitt et al., 2001). Information on the distribution of grass species among TDWG regions was obtained from GrassBase (Clayton et al., 2006 onward). This resulted in the inclusion of an additional 69 species, considered endemic (Supplementary Text, Supplementary Table S1), bringing the final total to 2,800 species (Supplementary Figure S1). The number of geographical occurrence records per species in this dataset ranged from

5 to just over 1 million (for *Holcus lanatus* L. and *Dactylis glomerata* L.).

Finally, to prevent a bias when linking the geographical data to climate data due to high sampling in certain areas, and to remove duplicate GBIF entries, the geographical dataset was reduced to one observation per tile (in a 30 arc-s grid) per species using the raster package (v. 3.1-5; Hijmans, 2020). Following this, the number of unique observations per species ranged from 1 to 175,810, averaging at 2,152 unique observations per species. The final dataset consisted of 2,800 species with both geographical and phylogenetic data and just over 6 million GBIF observations.

Defining areas experiencing drought, frost, and severe winter using Köppen–Geiger climate zones

It is notoriously difficult to define drought in an ecologically meaningful way, and to delimit areas that experience drought (Slette et al., 2019). Because of this, we used Köppen–Geiger climate zones to define geographical areas experiencing drought, frost, and severe winter (Supplementary Table S2). Köppen–Geiger zones divide the world into five main climate zones based on their vegetation composition: tropical (A), arid (B), temperate (C), continental (D), and polar (E), and each zone is then further divided into subzones based on temperature and precipitation criteria (Köppen, 2011; Beck et al., 2018). Each zone was scored as experiencing drought, frost, and/or winter as follows. Zones B (arid and semi-arid climates), Aw and As (tropical savannahs) were scored as experiencing drought. In addition, zone Csa was scored as experiencing drought due to its combination of high temperatures and low precipitation during the summer months (e.g., Los Angeles, Perth and Valencia).

For consistency, we also used Köppen–Geiger zones to define areas experiencing frost and severe winters (Supplementary Table S2). Köppen–Geiger zones classified as experiencing frost were defined as having one or more months averaging below 0 °C; the D and E zones fulfilled this criterion. Although zones Bxk, Cxb, and Cxc do not have any months averaging below 0 °C, they do experience occasional to regular frost during the winter months (e.g., Reno [Nevada, USA] and the Patagonian Desert for Bxk; Cape Town, Mexico City, and Plymouth [UK] for Cxb; and Balmaceda [Chile], Bodø [Norway] and the Faroe Islands for Cxc). Therefore, they were also scored as experiencing frost. Zones with severe winter were defined as those experiencing frost and a cool, short growing season with a maximum of three months averaging above 10 °C. This included zones Cxc, Dxc, Dxd, and E (Supplementary Table S2).

Species distribution thresholds and sensitivity analyses

Following the scoring of which geographical areas experience drought, frost, and severe winter, the GBIF geographical occurrence records were linked to this scoring to determine which species are exposed to drought, frost, or severe winter in their ranges using the R packages kgc (v. 1.0.0.2; Bryant et al., 2017) and raster (v. 3.1-5; Hijmans, 2020). We acknowledge that inferring physiological tolerances from occurrence patterns can both underestimate and overestimate true tolerances, e.g., if species distributions are strongly structured according to microclimatic variation (Greiser et al., 2020). However, it is still a useful approach at macroevolutionary scales (Humphreys & Linder, 2013).

We considered species drought, frost, and/or winter tolerant if at least 20% of their geographical observations fell in Köppen–Geiger zones scored accordingly. We chose this threshold to remove species occurring, but not established, in areas with drought, frost, or severe winter, with an emphasis on representing each species' optimal climate niche rather than its extremes. These species will hereafter be referred to as “drought tolerant,” “frost tolerant,” and “(severe) winter tolerant,” respectively. We also tested whether our results are contingent on the use of this threshold by creating two alternative datasets in which species were considered “tolerant” if at least 5% or 50% of their occurrence records fell in an area experiencing drought, frost, and/or winter. Analyses were run on data compiled using all three thresholds (20%, the “main” dataset; 5% and 50%, the “sensitivity” datasets), making nine datasets in total (3 traits × 3 thresholds).

Ancestral state reconstructions of drought, frost, and winter tolerance

To visualize the pattern of evolution for each of drought, frost, and winter tolerance separately, we performed ancestral state reconstructions using the “hidden rates” Markov models implemented in the R package corHMM (Beaulieu et al., 2013, 2020). This method takes rate heterogeneity during the course of evolution and among clades into account, and has been shown to perform similarly to other rate-variable methods (King & Lee, 2015). In the simplest corHMM model, the one-rate category model, a binary trait evolves with the same rate across the entire tree. This is equivalent to a standard model for a binary trait with one rate of gain and one rate of loss (all rates different; Beaulieu et al., 2013). The constraints on the model are then relaxed, stepwise, to generate increasingly complex models by adding additional rate categories. The two-rate category model thus allows each state to evolve in a fast and a slow rate category, with each rate category having its own rate of both gain and loss, as well as a rate of moving to the other rate category. In the three-rate category model, there are fast, medium, and slow rate categories, and so on, until the most complex model currently implemented in corHMM, the five-rate category model (Beaulieu et al., 2013). The models do not allow for both the trait and transition rate category to change simultaneously (e.g., from “sensitive-slow” to “tolerant-fast”), but multiple shifts of either type can happen along a single branch. The rate of changing from one rate category to another is independent of the trait (e.g., the rate of moving from the “fast” to the “slow” rate category is the same irrespective of whether the state is “sensitive” or “tolerant”). We fitted models with different numbers of transition rate categories, with the states “tolerant” and “sensitive,” across all nine datasets using maximum likelihood (ML) criteria and 100 random restarts. Model fit was assessed using Akaike's sample-size corrected information criterion (Burnham & Anderson, 2002; AICc). The estimated parameters of the best-fitting model were used to calculate the marginal probabilities of the most likely tolerant and sensitive state and rate at each internal node. Interpretations of model inferences were based on nodes with a marginal probability ≥ 0.75 for either tolerance or sensitivity (Beaulieu et al., 2013).

Testing for correlated evolution between drought and frost/winter tolerance

To test for correlated evolution between drought tolerance and each of frost and winter tolerance, we tested whether the

evolution of one trait (rate of gains and losses of frost and winter tolerance) is dependent on the state of the second trait (drought tolerance) by comparing the likelihood fit of two models (Pagel, 1994). The first model, the independent model, has four transition rate parameters, meaning that the rate of change in one trait is independent of the state of the second trait (Figure 3a). The second model, the correlated model, has eight possible rate parameters, meaning that the rate of change in one trait may differ depending on the state of the second trait (Figure 3b). If drought tolerance was a precursor for frost and/or winter tolerance, we would expect support for the correlated model, such that drought tolerance evolves first from the ancestral state (sensitive to drought, frost, and severe winter; as evidenced by transition rate 3 being higher than transition rate 1, Figure 3b) and transitions to frost and/or winter tolerance are more frequent in drought-tolerant than drought-sensitive lineages (as evidenced by transition rate 5 being higher than transition rate 1, Figure 3b).

We fitted the independent and correlated models using the one-rate category corHMM models as above for both trait combinations (drought-frost and drought-winter) and all three thresholds (six analyses in total). As there were no species with both drought and winter tolerance in the 50% threshold dataset, we fitted a reduced model with four-rate parameters to this dataset (Supplementary Text, Supplementary Figure S2a). Model fit was assessed using AICc.

As with several macroevolutionary methods, models of correlated evolution have been shown to have high type I error rates, i.e., rejection of a null hypothesis that is true (Boyko & Beaulieu, 2023; Maddison & FitzJohn, 2015; Rabosky & Goldberg, 2015). There can be several underlying causes of this, so we addressed this in two ways. First, we tested for evidence of correlated evolution using an alternative null model that takes rate heterogeneity into account. However, these models soon become very parameter rich and computationally demanding for a dataset of this size, meaning we may not be able to find the overall best model using this approach. Therefore, we also compared the strength of the evidence for correlated evolution in our drought tolerance data compared to data simulated to have evolved independently of either frost or winter tolerance.

Testing for correlated evolution using a null model with rate heterogeneity

False rejection of the null model may be because it is too trivial to serve as a meaningful null. This has been suggested for the independent model used here, specifically, because it fails to take rate heterogeneity into account (Boyko & Beaulieu, 2023). Thus, we compared the model of correlated evolution to an alternative null model: the independent model with two rate categories (the two-rate model). This model is formulated as two paired independent models, with four transition rates in each of the two rate categories (Supplementary Figure S2b). In addition, there are two transition rates for transitioning between rate categories. Thus, the two-rate independent model has ten transition rate parameters. We fitted the two-rate model across all six trait and threshold combinations using corHMM as before (Supplementary Text, Supplementary Figure S2b,c). Model fit was assessed using AICc.

Increasing model complexity further to fully address the question of correlated evolution, while adequately accounting for rate heterogeneity (Boyko & Beaulieu, 2023; King & Lee,

2015), would likely improve model fit, but such models soon become prohibitively complex and computationally demanding for a dataset of this size. This means the approach of comparing the correlated model to an alternative null model with rate heterogeneity can be used to test whether the initial rejection of the independent model could be due to statistical error, but not to find the best model overall. To address this technical challenge, we also assessed support for the correlated model using simulations.

Testing for correlated evolution using simulated data

We compared support for the correlated and independent models using data simulated to represent a binary trait that has evolved independently of either frost or winter tolerance. The rationale behind this approach is that no support for correlated evolution is expected in the simulated data. If there is, this points to statistical error (false rejection of the null model) and the difference in fit between independent and correlated models needs to be smaller for simulated than observed data for the correlated model still to be considered significantly supported for observed data. In other words, the simulations are used to find a new critical value for significance, given some level of type I statistical error (see also Humphreys & Barraclough, 2014; Humphreys et al., 2016; Rabosky & Goldberg, 2015).

We simulated 500 binary traits on the observed phylogenetic tree under a Markov model of evolution using the “sim.Mk” function in the R package phytools (Revell, 2012) and the transition rate matrix inferred from the best-fitting Markov model for drought tolerance (three-rate model; 20% dataset; Tables 1 and 2). Independent and correlated models with one transition rate category were then fitted to each simulated trait together with the observed frost and winter tolerance data in turn (20% threshold). Model fit was assessed using AICc.

To test whether the support for the correlated model found for the observed drought-frost and drought-winter data was greater than for the simulated data, we calculated the 95% confidence intervals (CIs) for the difference in fit between the independent and correlated models for simulated data, and compared this to the difference in fit between independent and correlated models for the observed data. The models of correlated evolution for observed data are considered supported if the difference in fit is significantly greater (>95% CI) than for simulated data. If it is not, this indicates that the original rejection of the null (independent) model in favor of the alternative (correlated) model is due to a type I statistical error.

Results

Distribution of sampled species and occurrence records among clades and Köppen–Geiger climate zones

Grasses have been recorded from every Köppen–Geiger zone. Of all filtered GBIF observations, 64% occurred in the *Cfb* zone (temperate oceanic climate), which covers large parts of Western Europe, New Zealand, and South-East Australia. The zones with the second and third most records were *Cfa* (humid subtropical climate) and *Dfc* (subarctic climate), accounting for only 5.8% and 5.5% of the observations, respectively. Of the 2,800 species in the dataset, 1,549 species occur in areas experiencing drought, 1,759 species occur in areas

experiencing frost, 375 species occur in areas with severe winters, 828 species are exposed to both drought and frost, and 57 species to both drought and severe winter (20% threshold dataset; [Supplementary Table S3](#)). The phylogeny includes around 25% of all Poaceae species, but species sampling is unevenly distributed among subfamilies ([Supplementary Text, Supplementary Table S4](#)), and GBIF records are unevenly distributed among species ([Supplementary Text, Supplementary Figure S3](#)).

Ancestral state reconstructions

We report the results for the 20% threshold dataset. Results for the 5% and 50% threshold datasets are summarized below. Under the 20% threshold dataset, *Pharus latifolius* L., sister to all other species sampled here, is scored as sensitive to drought, frost, and severe winter.

Drought tolerance

All major clades contain species scored as drought tolerant ([Figure 1, Supplementary Figure S4](#)), however, three clades stand out as having fewer drought-tolerant species: Pooideae (specifically Poeae), Bambusoideae (specifically Arundinarieae), and Danthonioideae (the clade sister to *Pentameris*). The two best-fitting models for drought tolerance were the three-rate and four-rate models ($\Delta\text{AICc} \geq 13.24$), which were statistically indistinguishable from each other ($\Delta\text{AICc} = 0.24$; [Table 1](#)). Therefore, we accepted the simpler three-rate model. Under this model, the ancestors of the PACMAD clade and all its constituent subfamilies were most likely drought tolerant ([Supplementary Figure S4](#)), with the highest marginal probabilities (≥ 0.75) for ancestral drought tolerance for PACMAD as a whole and all subfamilies except Panicoideae and Danthonioideae ([Figure 1](#)). However, several clades within Panicoideae (e.g., subtribe Boivinellinae) and Danthonioideae were reconstructed as being ancestrally drought tolerant ([Figure 1](#)). The reconstructions for the ancestors of the BOP clade and Poaceae as a whole were uncertain ([Figure 1, Supplementary Figure S4](#)). Within the BOP clade, no deeper nodes were reconstructed as being drought tolerant, but certain clades within Pooideae (e.g., *Aegilops* and within Stipeae), Bambusoideae, and Oryzoideae were reconstructed as drought tolerant. The ancestors of Pooideae (except *Brachyelytrum* that is sister to the rest) and Bambusoideae were most likely drought sensitive, as was the ancestor of Oryzoideae but with less certainty ([Figure 1, Supplementary Figure S4](#)).

Based on the sum of the gains and losses of tolerance, we identified a fast (R1), medium (R2), and slow (R3) rate category ([Table 2](#)). Drought tolerance in the fast and slow rate categories are reconstructed throughout the tree ([Supplementary Figure S4](#)): the fast category is reconstructed in all major clades, notably in much of Panicoideae, half of Danthonioideae (*Pentameris*), parts of Chloridoideae, Bambusoideae, Oryzoideae, and parts of Pooideae (tribes Stipeae and Triticeae); the slow rate category is primarily reconstructed in the Aristidoideae, Chloridoideae, a small number of lineages in Panicoideae (e.g., subtribe Boivinellinae), and *Aegilops* (Triticeae, Pooideae). Drought sensitivity is inferred mainly in the medium (R2) and slow (R3) rate categories, with the medium category being prevalent in the Pooideae (especially Poeae) and the slow category reconstructed in Bambusoideae (Arundinarieae) and Danthonioideae (sister clade to *Pentameris*).

Frost tolerance

Species belonging to all subfamilies were scored as being frost tolerant, with the Danthonioideae, Chloridoideae, and Pooideae having the most frost-tolerant species ([Figure 1, Supplementary Figure S4](#)). The three-rate category model was the best fit ($\Delta\text{AICc} \geq 11.06$; [Table 1](#)). Under this model, reconstructed ancestral states deeper in the tree are equivocal, but more certain for some of the subfamilies. Pooideae, Danthonioideae, and part of Chloridoideae (*Muhlenbergiinae sensu Peterson et al., 2010*) were most likely ancestrally frost tolerant and Bambusoideae and Panicoideae (excluding two small clades successively sister to the rest) most likely ancestrally frost sensitive (marginal probability ≥ 0.75 ; [Figure 1](#)).

Based on the sum of gains and losses, we distinguish fast (R1), medium (R3), and slow (R2) rate categories ([Table 2](#)). Frost tolerance in the medium rate category is inferred in the Pooideae, Oryzoideae (*Ehrharta*), Danthonioideae, and Chloridoideae (*Muhlenbergiinae sensu Peterson et al., 2010*; [Supplementary Figure S4](#)). Frost tolerance in the slow category is less frequently occurring and distributed more sporadically throughout the tree, notably being inferred in the Chloridoideae (*Zoysiaeae, Eragrostidinae*) and Panicoideae. Frost tolerance in the slow category is more easily lost than frost tolerance in the medium category (relative rate of loss 0.77 compared to 0.42, [Table 2](#)). For frost-sensitive clades, the fast rate category (R1) is most common but the slow category is often reconstructed equivocally at the same nodes as frost tolerance in the slow category ([Supplementary Figure S4](#)).

Severe winter tolerance

Species scored as tolerant of severe winter are clustered mainly in Pooideae and Danthonioideae, but a few winter-tolerant species occur also in other subfamilies ([Figure 2, Supplementary Figure S5](#)). The three-rate category model had the best fit for winter tolerance ($\Delta\text{AICc} \geq 3$; [Table 1](#)). Poaceae as a whole, the BOP and PACMAD clades, and each of the subfamilies were all inferred as being ancestrally winter sensitive (marginal probability ≥ 0.75 ; [Figure 2](#)). Very few internal nodes were reconstructed as winter tolerant, the exceptions being scattered clades within Danthonioideae (e.g., *Chionochloa*) and Pooideae (in Poeae, Triticeae, and Stipeae; marginal probability ≥ 0.75 ; [Figure 2](#)).

Based on the sum of gains and losses, we define the rate categories fast (R1), medium (R2), and slow (R3; [Table 2](#)). Winter tolerance is mostly found in the fast rate category ([Supplementary Figure S4](#)). This winter tolerance is often associated with, and evolves from, winter sensitivity in the medium and fast rate categories but never in the slow rate category. Lineages in the medium rate sensitive state are found in Pooideae and parts of Danthonioideae (sister to *Pentameris*) but also, notably, in *Muhlenbergiinae sensu Peterson et al. (2010)* (Chloridoideae) and the neotropical woody bamboos (Bambusoideae). Most other clades are not tolerant of severe winters and ancestral states are inferred to be sensitive in the slow category.

Models of correlated evolution between drought and frost/winter tolerance

The correlated model was a better fit than the independent model for both drought-frost and drought-winter ($\Delta\text{AICc} > 59$; [Table 3](#)). The patterns in these models are qualitatively

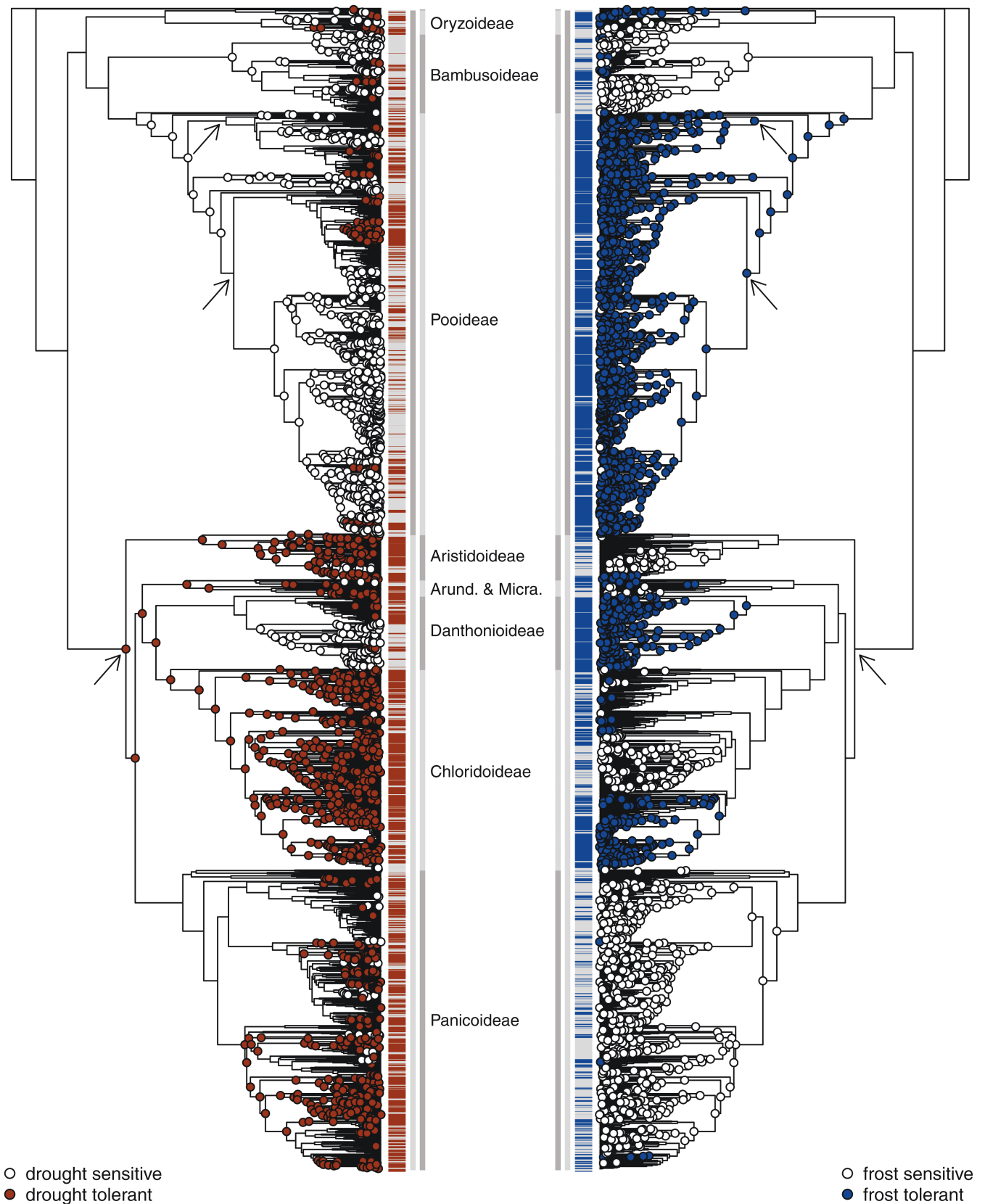


Figure 1. Ancestral state reconstructions for drought (left) and frost (right) tolerance inferred from the best-fitting hidden rates models (Tables 1 and 2). Ancestral states reconstructed with a marginal probability ≥ 0.75 are shown at each node, irrespective of rate category (filled = drought/frost tolerance and open = drought/frost sensitivity). Nodes that are unlabeled (no filled or open circle) were equivocal (marginal probability < 0.75 for either tolerant/sensitive). Tolerance/sensitivity of the sampled species is labeled at the tips (dark = drought/frost tolerance and light = drought/frost sensitivity). Outer vertical line: extent of labeled subfamilies ("Arund. & Micra." = Arundinoideae and Micrairoideae). Inner vertical line indicates the two major clades of grasses: dark: the BOP clade (Bambusoideae, Oryzoideae, and Pooideae); light: the PACMAD clade (Panicoideae, Aristidoideae, Chloridoideae,

Table 1. Likelihood and AICc scores for ancestral state reconstructions with different numbers of hidden rate categories (CorHMM) for each of the three datasets.

Number of rate categories	Drought			Frost			Winter		
20% threshold	Log-likelihood	AICc	ΔAICc	Log-likelihood	AICc	ΔAICc	Log-likelihood	AICc	ΔAICc
One-rate	-1,703.06	3,410.12	340.06	-1,450.15	2,904.31	415.81	-891.76	1,787.53	258.88
Two-rate	-1,542.86	3,097.75	27.69	-1,262.47	2,536.98	48.48	-778.66	1,569.36	40.71
Three-rate	-1,523.09	3,070.30	0.24	-1,232.19	2,488.50	0	-752.27	1,528.65	0
Four-rate	-1,514.88	3,070.06	0	-1,229.63	2,499.56	11.06	-745.91	1,532.13	3.48
Five-rate	-1,511.31	3,083.30	13.24	-1,227.03	2,514.74	26.24	-743.61	1,547.90	19.25
5% threshold									
One-rate	-1,479.34	2,962.68	347.24	-1,311.25	2,626.50	355.91	-1,116.55	2,237.10	326.34
Two-rate	-1,329.05	2,670.13	54.69	-1,143.39	2,298.80	28.21	-977.41	1,966.85	56.09
Three-rate	-1,295.66	2,615.44	0	-1,123.21	2,270.59	0	-943.41	1,910.93	0.17
Four-rate	-1,289.51	2,619.32	3.88	-1,120.57	2,281.45	10.86	-935.23	1,910.76	0
Five-rate	-1,288.19	2,637.06	21.62	-1,120.32	2,301.31	30.72	-931.14	1,922.96	12.20
50% threshold									
One-rate	-1,629.85	3,263.70	317.52	-1,388.07	2,780.17	388.47	-581.44	1,166.89	165.37
Two-rate	-1,467.73	2,947.50	1.32	-1,208.63	2,433.32	41.62	-499.67	1,015.40	13.88
Three-rate	-1,461.04	2,946.18	0	-1,185.64	2,399.43	7.73	-488.70	1,001.52	0
Four-rate	-1,456.42	2,953.14	6.96	-1,175.70	2,391.70	0	-482.10	1,004.51	2.99
Five-rate	-1,454.33	2,969.34	23.16	-1,171.75	2,404.18	12.48	-481.73	1,024.13	22.61

the same for frost and winter: drought tolerance is most likely to evolve first from the ancestral state of sensitivity to both drought and frost (higher rate of gain of drought tolerance than frost [0.14 vs 0.082] or winter [0.19 vs. 0.034] tolerance), but frost and winter tolerance are more likely to originate in drought-sensitive than drought-tolerant lineages (frost: 0.082 vs. 0.055; winter: 0.034 vs. 0.0056; Figure 3c,d).

The two-rate independent model was a much better fit than the single-rate correlated model for both drought-frost and drought-winter ($\Delta\text{AICc} > 234$; Table 3). Analysis of simulated data revealed that the null (independent) model was rejected in favor of the alternative (correlated) model for all simulated traits, indicating high type I error rates, but the difference in fit between the independent and correlated models was always much greater for the observed than simulated data for both drought-frost and drought-winter (Table 4).

Sensitivity analyses

The main results were the same across all datasets, with six exceptions (Supplementary Text; Tables 1 and 3, Supplementary Table S3). Four differences were for the 50% threshold datasets and two were for the 5% threshold datasets. The differences concerned scoring of drought tolerance for *Pharus latifolius*, scoring of severe winter tolerance in Bambusoideae, and model fit. None of these differences affected the overall reconstructed evolutionary scenario underlying the conclusions.

Discussion

We tested for a signature of evolutionary precursors in adaptation to a temperate climate. The importance of precursors in the formation of biomes and assembly of regional floras is known in the biogeographical literature as “biome conservatism” (Crisp et al., 2009) or the “temperate element” of high alpine floras (Gehrke & Linder, 2009; Gehrke, 2018; but see Nürk et al., 2018). Biome transitions require complex structural, physiological, and phenological adjustments, and it is reasonable to assume that not all such changes happened simultaneously. Instead, it is more plausible that present-day adaptations were acquired gradually, with the gradual accumulation of multiple new traits providing opportunities for persisting in one changing environment, as well as colonizing other emerging environments.

Despite some research (e.g., Edwards et al., 2017; Judd et al., 1994; Sakai & Larcher, 1987; Spriggs et al., 2015; Zanne et al., 2014), little is known, however, about the order in which adaptations to freezing and seasonal conditions originated in present-day temperate clades. Little is also known about the extent to which individual lineages still hold signatures of any ancient precursor traits that may have facilitated their biome transitioning journey. To address this, we used a comparative approach to test the idea that tropical-to-temperate transitions were facilitated by precursor trait(s). Specifically, we tested for evidence that drought tolerance acted as a precursor to frost and winter tolerance in grasses. That is, that drought tolerance evolved first and frost and winter tolerance originated more frequently in ancestrally drought-tolerant

Table 2. Rate matrix for the best-fitting hidden rates models (three-rate model) for the 20% threshold dataset.

a) Drought tolerance							
Rate	Rate	Fast (R1)		Medium (R2)		Slow (R3)	
	Tolerance	Sensitive	Tolerant	Sensitive	Tolerant	Sensitive	Tolerant
Fast (R1)	Sensitive	–	100.00	0.015	–	0.029	–
	Tolerant	40.76	–	–	0.015	–	0.029
Medium (R2)	Sensitive	0.033	–	–	7.81	0.020	–
	Tolerant	–	0.033	40.76	–	–	0.020
Slow (R3)	Sensitive	0.037	–	0.002	–	–	0.000000001
	Tolerant	–	0.037	–	0.002	0.000000001	–
b) Frost tolerance							
Rate	Rate	Fast (R1)		Slow (R2)		Medium (R3)	
	Tolerance	Sensitive	Tolerant	Sensitive	Tolerant	Sensitive	Tolerant
Fast (R1)	Sensitive	–	2.77	0.023	–	0.000000001	–
	Tolerant	13.21	–	–	0.023	–	0.000000001
Slow (R2)	Sensitive	0.04	–	–	1.78	0.016	–
	Tolerant	–	0.04	0.77	–	–	0.016
Medium (R3)	Sensitive	0.000000001	–	0.009	–	–	8.78
	Tolerant	–	0.000000001	–	0.009	0.42	–
c) Winter tolerance							
Rate	Rate	Fast (R1)		Medium (R2)		Slow (R3)	
	Tolerance	Sensitive	Tolerant	Sensitive	Tolerant	Sensitive	Tolerant
Fast (R1)	Sensitive	–	100.00	0.09	–	0.019	–
	Tolerant	50.66	–	–	0.09	–	0.019
Medium (R2)	Sensitive	0.028	–	–	1.35	0.021	–
	Tolerant	–	0.028	14.23	–	–	0.021
Slow (R3)	Sensitive	0.000000001	–	0.007	–	–	0.000000001
	Tolerant	–	0.000000001	–	0.007	0.024	–

lineages, suggesting that ancestral drought tolerance facilitated transitions into cold temperate environments (Preston & Sandve, 2013; Sakai & Larcher, 1987).

Ancestral state reconstructions reveal both drought-first and frost-first scenarios

Drought tolerance is inferred to have evolved early in the PACMAD clade, with the ancestor most likely being drought tolerant (Figure 1; Supplementary Figure S4). This is consistent with drought tolerance evolving simultaneously with the transition from closed to open environments in the ancestor of the PACMAD clade (Kellogg, 2001; Strömberg, 2011; Elliott et al., 2023). In contrast, drought tolerance was not reconstructed deep in the BOP clade, consistent with multiple independent transitions to open habitats in this clade, although our reconstruction of drought tolerance suggests it evolved long after transitions to open environments occurred (Kellogg, 2001; Zhang et al., 2022a; Elliott et al., 2023). The relatively late evolution of drought tolerance in Pooideae is consistent with the “tippy” distribution of xerophytes and high evolutionary lability of drought responses found previously for this clade (Zhang et al., 2022a; Pal Stolsmo et al., 2024).

Overall, our reconstruction suggested a later origin of frost tolerance than drought tolerance (Figure 1; Supplementary Figure S4). Frost tolerance was not inferred as ancestral in either the BOP or PACMAD clades, but the ancestors of Pooideae, Danthonioideae, and part of Chloridoideae (*Muhlenbergiinae*; Peterson et al., 2018) were reconstructed as frost tolerant. The overall deeper origin of drought than frost tolerance supports the view that grasses were exposed to drought before frost during their evolutionary history (Linder et al., 2018; Strömberg, 2011). Deep origins of frost tolerance in Pooideae and Danthonioideae is also consistent with previous studies (Humphreys & Linder, 2013; Schubert, Marcussen & et al., 2019). However, upon scrutiny, it is clear that frost-tolerant lineages in the BOP and PACMAD clades have undergone different evolutionary trajectories; drought tolerance most likely having preceded frost tolerance in Danthonioideae and Chloridoideae but not in Pooideae. In other words, frost tolerance evolved in ancestrally drought-tolerant lineages in Danthonioideae and Chloridoideae, but in ancestrally drought-sensitive lineages in Pooideae.

Tolerance of severe winters was reconstructed ancestrally for certain clades within Danthonioideae (especially *Chionochloa*) and Pooideae (within tribes Poeae, Triticeae,

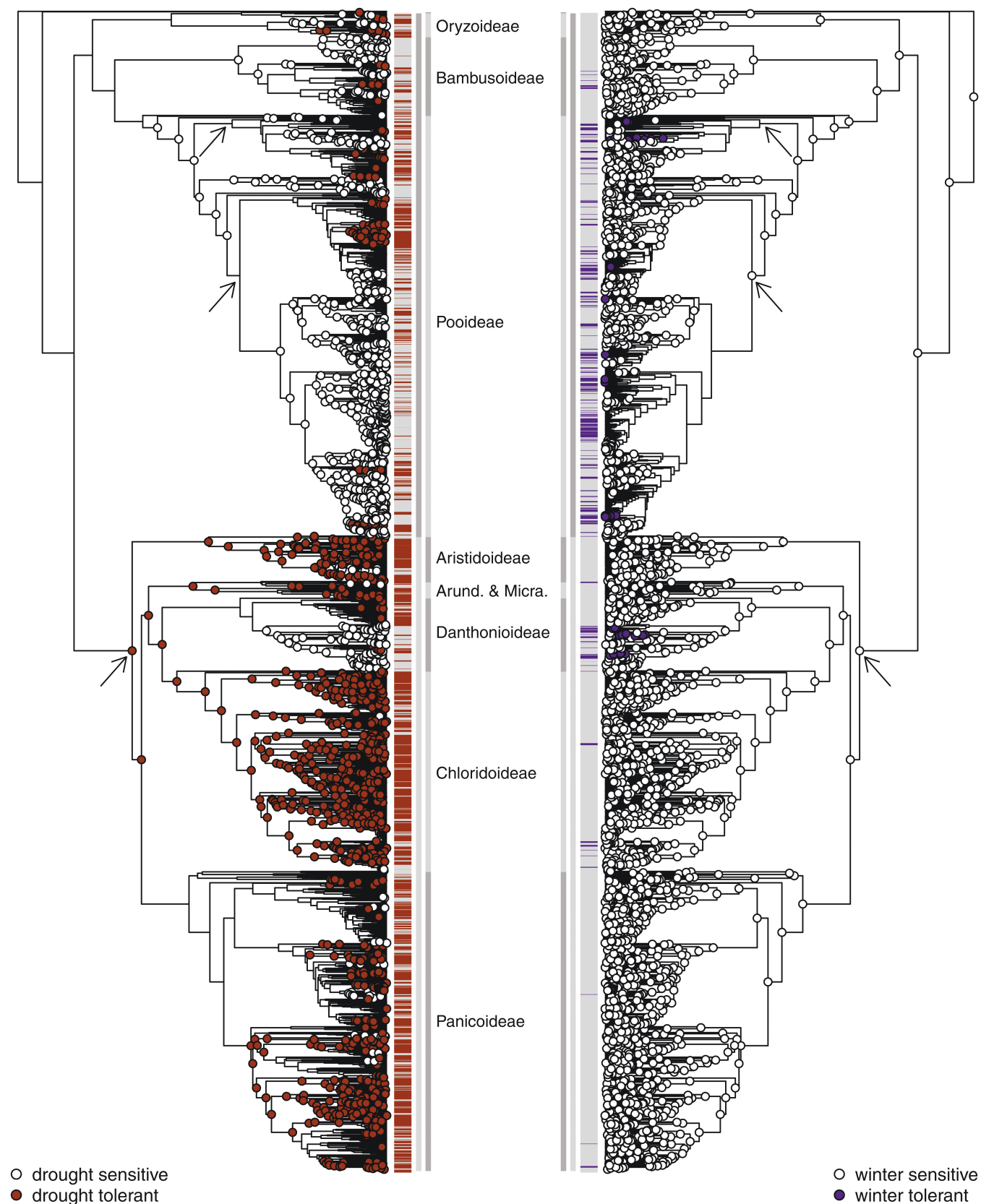


Figure 2. Ancestral state reconstructions for drought (left) and severe winter (right) tolerance inferred from the best-fitting hidden rates model (Tables 1 and 2). Ancestral states reconstructed with a marginal probability ≥ 0.75 are shown at each node, irrespective of rate category (filled = drought/winter tolerance, and open = drought/winter sensitivity). Nodes that are unlabeled (no filled or open circle) were equivocal (marginal probability < 0.75 for either tolerance/sensitivity). Tolerance/sensitivity of the sampled species is labeled at the tips (dark = drought/winter tolerance and light = drought/winter sensitivity). Outer vertical line: extent of labeled subfamilies ("Arund. & Micra." = Arundinoideae and Micraioideae). Inner vertical line indicates the two major clades of grasses; dark: the BOP clade (Bambusoideae, Oryzoideae, and Pooideae); light: the PACMAD clade (Panicoideae, Aristidoideae,

and Stipeae; Figure 2). In addition, scattered occurrences of winter tolerance were found in Bambusoideae, Chloridoideae, and Panicoideae. In all these clades except Panicoideae, winter tolerance is preceded by winter sensitivity in the medium rate category (Supplementary Figure S5), suggesting that this state may represent a precursor trait (but see Supplementary Text for the reconstruction in Bambusoideae). Tolerance of severe winters encompasses both physiological and phenological adaptations, and a plausible prerequisite is adaptation to periodic (seasonal) frost. Our results of the ancestral state reconstructions corroborate this, at least for Pooideae and parts of Danthonioideae and Chloridoideae (Supplementary Text, Supplementary Figure S5). Since we found patterns consistent with drought tolerance as a precursor to frost tolerance in the latter two clades, drought tolerance could indirectly be a precursor to severe winter tolerance in those clades as well.

Models of correlated evolution supported—with important caveats

The above discussion is based on comparison of separate analyses for each of drought, frost, and winter tolerance. The explicitly simultaneous modelling of drought-frost and drought-winter tolerance showed that their evolution has been correlated, that drought tolerance is the most likely to originate first from the ancestral state, but that both frost and winter tolerance most likely originated in drought-sensitive rather than drought-tolerant lineages (Table 3, Figure 3c,d). These models were supported by the analyses of simulated data, which showed that the difference in fit between independent and correlated models for observed data greatly exceeds that for simulated data (Table 4). This demonstrates that the signal of correlated evolution is significantly stronger than expected for a trait evolving independently of either frost or winter tolerance. Taken at face value, these results support the overall deeper origin of drought tolerance observed in the ancestral state reconstructions, but suggest that evolution of drought and frost/winter tolerance have been negatively correlated, implying that drought tolerance is unlikely to have acted as an evolutionary precursor to either frost or winter tolerance.

However, there are two caveats to this interpretation. The first is technical. Comparison of the fit of the correlated models against alternative null models that allow for rate heterogeneity revealed that incorporating rate heterogeneity leads to much better models (Table 3). Increasing model complexity even further to test for correlated evolution while accounting for rate heterogeneity (Boyko & Beaulieu, 2023; King & Lee, 2015) becomes prohibitively complex and computationally demanding for a dataset of this size. This means we are unlikely to have found the best models for evolution of drought-frost or drought-winter, we have only demonstrated that there is support for correlated evolution—in the absence of rate heterogeneity.

The second caveat is biological. Our ancestral state reconstructions reveal different evolutionary trajectories in different clades (Figures 1 and 2). Modelling the Poaceae as

a whole may have obscured these different trajectories. A straightforward solution would be to analyze the BOP and PACMAD clades separately; however, our results point to a single deep origin of drought tolerance in the PACMAD clade, followed by repeated origins of frost tolerance. This scenario thus reduces to a single evolutionary event that lacks the comparison of ancestrally drought-sensitive clades (Maddison & FitzJohn, 2015; Uyeda et al., 2018). In addition, analysis of smaller clades carries the risk of incorrect interpretations, if they are not representative of the broader group of interest (Beaulieu & O'Meara, 2018). This leaves the option of addressing the question at even broader phylogenetic scales (e.g., the Poales), but this approach is marred by the challenge of accurately modelling the many diverse histories that are inevitable at such broad scales. Thus, in lieu of more complex modelling in carefully (Donoghue & Edwards, 2014) or randomly (Beaulieu & O'Meara, 2018) selected clades, i.e., ones that include the complexity required to test the hypothesis without being too large to fit the models that can adequately explain that complexity, our conclusion is that drought tolerance probably did originate before either frost or winter tolerance in grasses overall. However, drought tolerance did not necessarily act as an evolutionary precursor to frost or winter tolerance, since the ancestral state reconstruction revealed different evolutionary scenarios in different clades and the overall best model tested did not support correlated evolution.

Conclusions

Our findings provide new insight into how drought, frost, and winter tolerance have evolved relative to each other at macroevolutionary scales. The ancestral state reconstructions revealed two different evolutionary trajectories for drought, frost, and winter tolerance in grasses. Drought tolerance most likely evolved first in the PACMAD clade, followed by repeated entries into freezing climates, followed in Danthonioideae by adaptation to severe winters. In the BOP clade, frost tolerance most likely evolved first, followed by adaptation to severe winters in some clades and dry environments in others. The drought-first scenario for the PACMAD clade is consistent with our expectations and evidence that grasses were exposed to dry before freezing conditions (e.g., Linder et al., 2018; Strömberg, 2011). In light of this, the frost-first scenario for the BOP clade is surprising, but consistent with recent evidence from ecophysiological experiments and gene expression patterns in Pooideae (Das et al., 2023; Pal Stolsmo et al., 2024, as well as suggestions that Pooideae originated in emerging cool climate pockets present as early as the Late Cretaceous (Schubert et al., 2020). Therefore, our results add to mounting evidence that frost tolerance evolved before drought tolerance in the BOP clade. Alternatively, it is possible that the signature of the shared ancient pathways giving rise to modern-day stress responses is no longer detectable in some clades.

The different evolutionary trajectories in the BOP and PACMAD clades may also explain the lack of support for

Chloridoideae, Micrairoideae, Arundinoideae, and Danthonioideae). The two successive sisters to the rest sampled here (*Puelia olyrififormis* and *Pharus latifolius*) are not part of any labeled subfamily or clade. Arrows indicate ancestral nodes where transitions from closed to open habitats may have occurred (Kellogg, 2001; Bouchenak-Khelladi et al., 2010; Zhang et al., 2022a; Elliott et al., 2023). The detailed reconstructions showing all states in different rate categories are shown in Supplementary Figure S5.

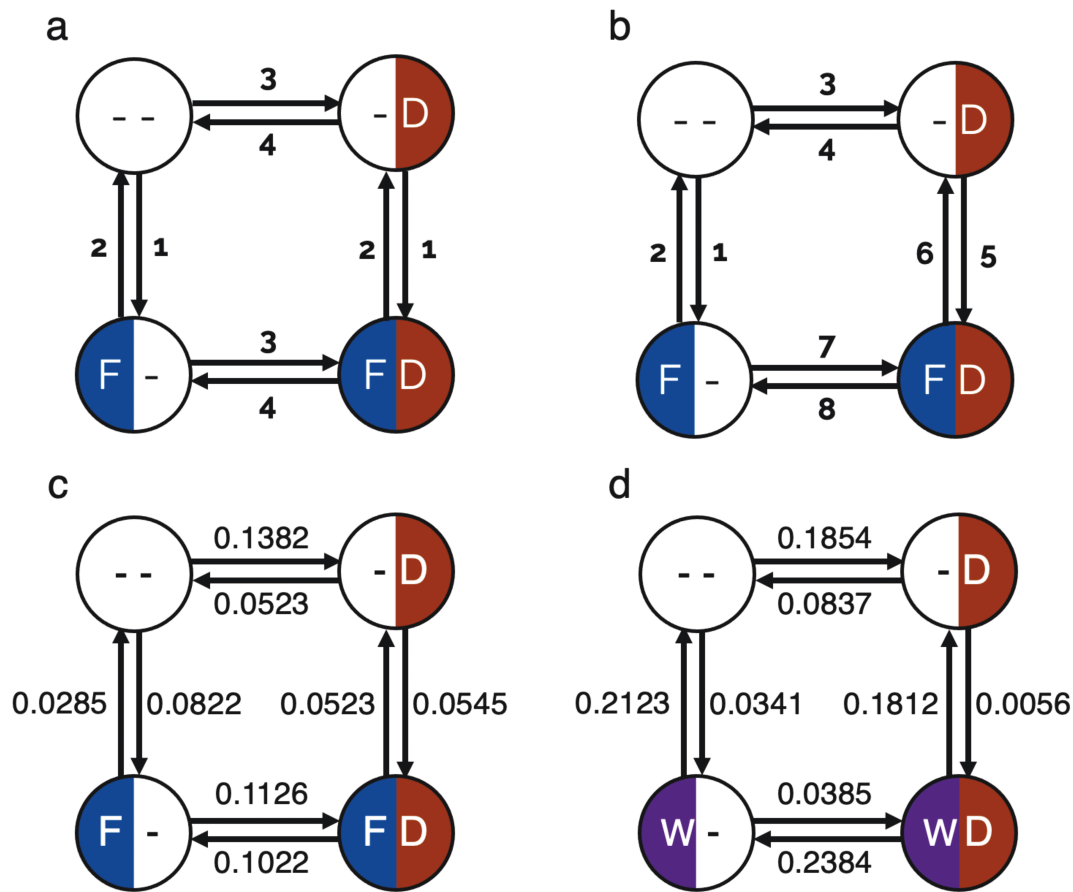


Figure 3. Schematic indicating states and transition parameters of the models used in the study: the (a) single-rate independent and (b) single-rate-correlated models, and the inferred transition rates based on the best-fitting single-rate models of correlated evolution for (c) drought and frost tolerance and (d) drought and severe winter tolerance. The circles represent the four different character state combinations: “-” = sensitivity to drought/frost/winter, “D” = drought tolerance, “F” = frost tolerance, and “W” = winter tolerance. The transition rates associated with each arrow represent the rates of gains and losses of drought, frost, and winter tolerance. For (a,b), frost tolerance (F) can be substituted for severe winter tolerance (W).

Table 3. Likelihood and AICc scores for the one-rate independent, one-rate correlated, and two-rate independent models for each of the six datasets (three thresholds and two trait combinations).

Number of rate cats.	Frost-drought			Winter-drought		
	-lnL	AICc	ΔAICc	-lnL	AICc	ΔAICc
20% threshold						
One-rate independent	-3,153.21	6,314.43	414.25	-2,594.82	5,197.65	403.66
One-rate correlated	-3,119.91	6,255.89	355.71	-2,506.20	5,028.46	234.47
Two-rate independent	-2,940.05	5,900.18	0	-2,386.95	4,793.99	0
5% threshold						
One-rate independent	-2,790.59	5,589.19	357.81	-2,595.89	5,199.79	464.14
One-rate correlated	-2,781.61	5,579.27	347.89	-2,520.76	5,057.56	321.91
Two-rate independent	-2,605.65	5,231.38	0	-2,357.79	4,735.65	0
50% threshold						
One-rate independent	-3,017.92	6,043.86	482.34	-2,141.79 ^a	4,291.59 ^a	320.87 ^a
One-rate correlated	-2,910.24	5,836.54	275.02	NA ^a	NA ^a	NA ^a
Two-rate independent	-2,770.72	5,561.52	0	-1,975.32	3,970.72	0

^aFor winter-drought based on the 50% threshold, we ran a reduced one-rate model instead of the full independent and correlated models (see [Supplementary Text](#)).

Table 4 Difference in fit between one-rate independent and correlated models of evolution for simulated and observed data.

	ΔAICc — drought-frost	ΔAICc — drought-winter
Simulated data (95% confidence interval across 500 traits)	16.02–19.46	16.16–19.97
Observed data	58.54	169.19

correlated evolution for drought-frost or drought-winter, when rate heterogeneity is taken into account. However, the difficulties of fitting complex models to large datasets mean that we are unlikely to have found the best models describing evolution of these traits simultaneously. We conclude that drought tolerance originated before either frost or winter tolerance in grasses overall but did not necessarily act as an evolutionary precursor to frost or winter tolerance, as evidenced by the different reconstructed evolutionary scenarios in different clades, and the lack of support for correlated evolution in the best model among the set tested.

Interestingly, the different reconstructed evolutionary trajectories reveal a mirrored pattern, with clades occupying either predominantly dry or very cold environments. This mirrored pattern can be seen throughout the grasses and most likely underlies the finding that frost and winter tolerance originated in drought-sensitive lineages. These results are consistent with modern-day grasses being either drought or cold specialists (Visser et al., 2013), suggesting the existence of a trade-off, such that species tend not to be equally well adapted to both drought and cold, as has been found for woody temperate plants (Puglielli et al., 2021).

Finally, what is the consequence of relying on geographical occurrences instead of inherent physiological tolerances? While the relative tolerances of species are likely related to their distributions (Humphreys & Linder, 2013), absence from a region experiencing drought, frost, or severe winter, as defined here, does not necessarily equate to an inability to withstand those stresses. As such, our method may be underestimating the prevalence of each adaptation across the family. Furthermore, if species distributions follow microclimatic variation, our reliance on coarse-grid occurrence records could both underestimate and overestimate the prevalence of drought- and frost-tolerant species. Severe winter tolerance, as defined here, should be less affected by microclimatic specializations as seasonal cold and frost coupled with short growing seasons occurs across large, high-latitude areas. Importantly, our findings for Pooideae corroborate recent findings based on measured ecophysiological responses to frost and drought exposure (Pal Stolsmo et al., 2024), capture patterns of previous ancestral state reconstructions for grasses (e.g., Edwards & Smith, 2010; Humphreys & Linder, 2013; Watcharamongkol et al., 2018; Schubert et al., 2019, 2020; Zhang et al., 2022a; Elliott et al., 2023) and are robust to alternative treatments of the data (distribution thresholds; Supplementary Text). Thus, while the concerns regarding reliance on distribution data are real, they are unlikely to be biasing our results here.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

All data needed to reproduce this study are available from (Schat et al., 2024), via <https://doi.org/10.5061/dryad.7wm37pw3s>.

Author contributions

L.S., M.S., S.F., and A.M.H. designed the study. L.S. compiled the data, L.S. and M.S. performed the analyses, and all authors contributed to the interpretation of the results. L.S. and A.M.H. wrote the manuscript with input from M.S. and S.F.

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Conflict of interest: The authors declare no conflict of interest.

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